

The Hamiltonian cycle passing through prescribed edges in hypercubes with adjacently faulty vertices *

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Abstract

In this paper, we consider the problem of a fault-free Hamiltonian cycle passing through prescribed edges in hypercube Q_n with some adjacently faulty vertices. Let $F_e \subset E(Q_n)$ be a set of faulty edges, F_a be a set of adjacently faulty vertices, $E_0 \subset (E(Q_n) - F_e - F_a)$ be the set of prescribed edges where the subgraph induced by E_0 is a linear forest (i.e., pairwise vertex-disjoint paths). We construct a Hamiltonian cycle of $Q_n - F_e - F_a$ passing through all edges of E_0 for any $|E_0| \leq 2n - 4 - 2|F_e| - 2|F_a|$ and $|F_e| + |F_a| \leq n - 2$.

1 Introduction

The n -dimensional hypercube Q_n is one of the most popular and efficient interconnection networks. It possesses many excellent properties such as recursive structure, symmetry, small diameter, low degree, popular structure embedding, easy routing, and optimal connectivity and fault-tolerance. There are a lot of literatures for fault-tolerance about hypercube. The Hamiltonian property is important for the topology of interconnection networks. When we construct a fault-free Hamiltonian cycle of some interconnection network, we can simulate various algorithms on rings efficiently. It is well known that the hypercube is a Hamiltonian graph. In this paper, we will investigate some fault-tolerance property for Hamiltonian cycle of hypercube.

The interconnection network is usually represented by an undirected graph where vertices represent processors and edges represent links. Let $G = (V_0 \cup V_1, E)$ be a bipartite graph where V_0 and V_1 are two disjoint vertex sets such that each edge

of E consists of one vertex from each set. A *Hamiltonian path* is a path that spans G . A *Hamiltonian cycle* is a cycle that visits every vertex exactly once. A *Hamiltonian graph* is a graph that contains a Hamiltonian cycle. A graph $G = (V, E)$ is *k edges Hamiltonian* if $G - F_e$ is a Hamiltonian graph for every $F_e \subset E$ and $|F_e| \leq k$. A bipartite graph $G = (V_0 \cup V_1, E)$ is *Hamiltonian laceable* if there exists a *Hamiltonian path* between x, y for any $x \in V_0, y \in V_1$. A bipartite graph $G = (V_0 \cup V_1, E)$ is *k edges Hamiltonian laceable* if $G - F_e$ is Hamiltonian laceable for every $F_e \subset E$ and $|F_e| \leq k$. Tsai et al. proved that Q_n is $(n - 2)$ edges Hamiltonian laceable in [6]. Hung et al. investigated the adjacent vertices fault tolerance Hamiltonian laceable for hypercube. Let F_e be the set of faulty edges and F_a be the set of adjacently faulty vertices of hypercube. We use $|F_e|$ denote the number of faulty edges of F_e and $|F_a|$ the number of pairs of adjacently faulty vertices of F_a . A bipartite graph $G = (V_0 \cup V_1, E)$ is $|F_a|$ -adjacency F_e edges *Hamiltonian* if $G - F_a$ is f_e edges Hamiltonian. A bipartite graph $G = (V_0 \cup V_1, E)$ is $|F_a|$ -adjacency F_e edges *Hamiltonian laceable* if $G - F_a$ is F_e edges Hamiltonian laceable. They proved that Q_n is $|F_a|$ -adjacency $(n - 2 - |F_a|)$ edges Hamiltonian for $|F_a| \leq (n - 2)$ and $|F_a|$ -adjacency $(n - 2 - |F_a|)$ edges Hamiltonian laceable for $|F_a| \leq (n - 3)$ in [1].

On the other hand, Dvořák investigated the problem about prescribed edges. Given a set of fault-free prescribed edges in a hypercube, is there a Hamiltonian cycle passing through all edges of this set in the hypercube? In [2], Dvořák showed that Q_n contains a Hamiltonian cycle passing through all edges of any set of prescribed edges E_0 if and only if the subgraph induced by E_0 consists of pairwise vertex-disjoint paths and $|E_0| \leq 2n - 3$. Wang et al. further presented the result that all edges of E_0 lie on a Hamiltonian cycle of $Q_n - F_e$

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if E_0 is a linear forest (i.e., pairwise vertex-disjoint paths) and $|F_e| \leq n - 2$, $|E_0| \leq 2n - 3 - 2|F_e|$ in [7]. In this paper, we will investigate the adjacent vertices fault-tolerance for this problem. We will show that all edges of E_0 lie on a Hamiltonian cycle of $Q_n - F_a - F_e$, if $|F_e| + |F_a| \leq n - 2$, $|E_0| \leq 2n - 4 - 2|F_e| - 2|F_a|$ where the subgraph induced by E_0 is a linear forest.

2 Preliminaries

In this section, we will introduce some notations and properties which are required for our results.

An n -dimensional hypercube $Q_n = (V_0 \cup V_1, E)$ is a bipartite graph with 2^n vertices. The vertices are assigned binary strings of length n ranging from 0 to $2^n - 1$. Two vertices are adjacent if they differ only in one bit position. Let u_i be the i bit of the vertex u . If the two binary strings of u and v differ in the i th bit position, where $1 \leq i \leq n$, then the edge e is called a dimension i edge of Q_n . Let i be a fixed position, we use Q_{n-1}^0 to denote the subgraph of Q_n induced by $\{u \in V(Q_n) | u_i = 0\}$ and Q_{n-1}^1 to denote the subgraph of Q_n induced by $\{u \in V(Q_n) | u_i = 1\}$. We say that Q_n is decomposed into Q_{n-1}^0 and Q_{n-1}^1 by dimension i , and Q_{n-1}^0 and Q_{n-1}^1 are $(n - 1)$ -dimensional subcube of Q_n induced by the vertices with the i th bit position being 0 and 1, respectively. The graphs Q_{n-1}^0 and Q_{n-1}^1 are all isomorphic to Q_{n-1} . For each vertex $u \in V(Q_{n-1}^0)$, there is exactly one vertex in Q_{n-1}^1 , denoted by $u^{(1)}$, such that $(u, u^{(1)}) \in E(Q_n)$. Conversely, for each $u \in V(Q_{n-1}^1)$, there is one vertex in Q_{n-1}^0 , denoted by $u^{(0)}$, such that $(u, u^{(0)}) \in E(Q_n)$.

A *path* is a sequence of adjacent vertices, denoted as $\langle v_1, v_2, \dots, v_n \rangle$, where all vertices are distinct. We also denote this path as $\langle v_1, v_2, \dots, v_i \xrightarrow{P(v_i, v_j)} v_j, \dots, v_n \rangle$ where $P(v_i, v_j)$ is the subpath $\langle v_i, v_{i+1}, \dots, v_j \rangle$. A *cycle*, written as $\langle v_1, v_2, \dots, v_n \rangle$, is a path for $v_1 = v_n$. Let $d(u, v)$ be the distance of vertices u and v .

Let F_e and F_a be the set of faulty edges and adjacently faulty vertices, respectively. Let E_0 be the set of prescribed edges for $E_0 \subset E(Q_n - F_e - F_a)$. Let F_a^i, F_e^i, E_0^i be the set of adjacently faulty set, faulty edges, prescribed edges of Q_{n-1}^i , respectively, for $i = 0, 1$. Let $|F_e^i|$ and $|E_0^i|$ be the number of edges of F_e^i and E_0^i , respectively, and $|F_a^i|$ be the number of pairs of adjacently faulty

vertices in F_a^i , for $i = 0, 1$.

The lemmas in the follows are required for our proofs. The authors of [1] introduced the following lemma.

Lemma 1 *The graph Q_n is f -adjacency $(n - 2 - f)$ edges Hamiltonian cycle for $f \leq (n - 2)$, f -adjacency $(n - 2 - f)$ edges Hamiltonian laceable for $f \leq (n - 3)$.*

Some literatures showed that there exist two spanning disjoint paths between any two pairs of vertices with odd distance of hypercubes. The following lemma is the version proved in [2].

Lemma 2 *Let u, v, x, y be pairwise distinct vertices of Q_n , $n \geq 2$, and such that both $d(u, v)$ and $d(x, y)$ is odd. Then*

- (i) *there exist paths between u and v and between x and y whose vertex sets partition $V(Q_n)$,*
- (ii) *if $d(x, y) = 1$, there exists a Hamiltonian path of $Q_n - \{x, y\}$ between u and v unless $n = 3$, $d(u, v) = 1$ and $d(\{u, v\}, \{x, y\}) = 2$.*

The following result is showed in [7].

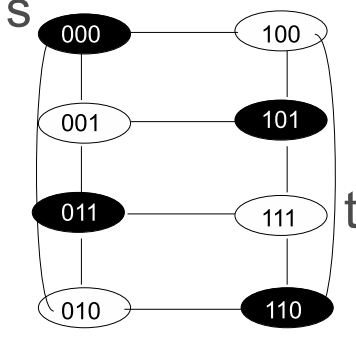
Lemma 3 *Let $n \geq 2$, $F \subset E(Q_n)$, $E_0 \subset E(Q_n) \setminus F$ with $1 \leq |E_0| \leq 2n - 3$, $|F| \leq n - 2$. If the subgraph induced by E_0 is a linear forest, then in the graph $Q_n - F$ all edges of E_0 lie on a Hamiltonian cycle.*

In the follows, we will construct a Hamiltonian path passing through a prescribed edge between every pair of vertices with odd distance of Q_n .

Lemma 4 *There exists a Hamiltonian path passing through any prescribed edge $e_0 = (u, v)$ between any two vertices s and t with $s \in V_0$ and $t \in V_1$ with $\{u, v\} \neq \{s, t\}$.*

Proof.

We will prove this lemma by induction on n . This lemma is obviously true for $n = 1, 2$. Let e_0 be the prescribed edge. We will first prove the case for $n = 3$. Suppose that $d(s, t) = 1$. Applying Lemma 3, we can obtain a Hamiltonian cycle passing through (s, t) and e_0 . Thus, there exists a Hamiltonian path $P(s, t)$ passing through e_0 . In the follows, we will consider the case for $d(s, t) \geq 2$. Thus, $d(s, t) = 3$. Without loss of generality, we can assume that $s = 000$ and $t = 111$, as illustrated in Figure 1. We can

Figure 1: The base case of Lemma 4 for $n = 3$.

construct the four Hamiltonian paths of Q_3 :

- (1). $\langle 000, 100, 101, 001, 011, 010, 110, 111 \rangle$;
- (2). $\langle 000, 100, 110, 010, 011, 001, 101, 111 \rangle$;
- (3). $\langle 000, 010, 110, 100, 101, 001, 011, 111 \rangle$;
- (4). $\langle 000, 001, 011, 010, 110, 100, 101, 111 \rangle$.

Since every edge of Q_3 is on at least one of these four Hamiltonian paths, this lemma holds for $n = 3$.

In the follows, we will prove this lemma for $n \geq 4$. Without loss of generality, we can assume the $e_0 \in Q_{n-1}^0$ or $e_0 \in Q_{n-1}^1$. We will show this lemma for $n \geq 4$ in the following cases.

Case1. s, t in the same subcube

Without loss of generality, we can assume that $s, t \in Q_{n-1}^0$.

Case1.1. $e_0 \in Q_{n-1}^0$

By induction hypothesis, this lemma holds for Q_{n-1} . There exists a Hamiltonian path $P(s, t)$ passing through the edge e_0 of Q_{n-1}^0 . Let (x, y) be an edge of $P(s, t)$ with $(x, y) \neq e_0$. Without loss of generality, we can assume that $P(s, t)$ as $\langle s \xrightarrow{P(s,y)} y, x \xrightarrow{P(x,t)} t \rangle$. Applying Lemma 1, we can obtain that there exists a Hamiltonian paths $P(y^{(1)}, x^{(1)})$ in Q_{n-1}^1 . Thus, $\langle s \xrightarrow{P(s,y)} y, y^{(1)} \xrightarrow{P(y^{(1)}, x^{(1)})} x^{(1)}, x \xrightarrow{P(x,t)} t \rangle$ forms a Hamiltonian path passing through e_0 of Q_n .

Case1.2. $e_0 \in Q_{n-1}^1$

Let $x \in V_1, y \in V_0$ be two vertices of Q_{n-1}^0 for $\{x, y\} \cap \{s, t\} = \emptyset$. Applying Lemma 2, we can obtain that there exist two spanning disjoint paths $P(s, x)$

and $P(y, t)$ of Q_{n-1}^0 . By induction hypothesis, there exists a Hamiltonian path $P(x^{(1)}, y^{(1)})$ passing through e_0 of Q_{n-1}^1 . Thus, $\langle s \xrightarrow{P(s,x)} x, x^{(1)} \xrightarrow{P(x^{(1)}, y^{(1)})} y^{(1)}, y \xrightarrow{P(y,t)} t \rangle$ is a Hamiltonian path passing through e_0 of Q_n .

Case2. s, t in different subcube

Without loss of generality, we can assume the $s \in Q_{n-1}^0, t \in Q_{n-1}^1$ and $e_0 \in Q_{n-1}^0$. Let $x \in V_1$ be a vertex of Q_{n-1}^0 . By induction hypothesis, there exist a Hamiltonian path $P(s, x)$ passing through e_0 of Q_{n-1}^0 and a Hamiltonian path $P(x^{(1)}, t)$ of Q_{n-1}^1 . Thus, $\langle s \xrightarrow{P(s,x)} x, x^{(1)} \xrightarrow{P(x^{(1)}, t)} t \rangle$ is a Hamiltonian path passing through e_0 of Q_n . $\square$

3 Main result

In this section, we will show the main result of this paper.

Theorem 1 *All edges of E_0 lie on a Hamiltonian cycle of $Q_n - F_a - F_e$, if $|F_e| + |F_a| \leq n - 2$, $|E_0| \leq 2n - 4 - 2|F_e| - 2|F_a|$ where the subgraph induced E_0 is linear forest.*

Proof. We will prove this theorem by induction on n . This theorem is true for $|F_a| = 0$ with Lemma 3. When $|F_e| + |F_a| = n - 2$, E_0 is an empty set. Thus this theorem is true for $|F_e| + |F_a| = n - 2$ with Lemma 1. In the follows, we will suppose that $|F_a| \geq 1$ and $|F_e| + |F_a| \leq n - 3$.

At first we will prove the base case for $n \leq 4$. This theorem is obviously true for $n \leq 3$. We only need to show the case for $|F_a| = 1$ and $|E_0| = 2$ for $n = 4$. With the symmetry of hypercube, we can assume that every edge of F_a or E_0 is either in Q_{n-1}^0 or Q_{n-1}^1 . Let $F_a = \{x, y\}$ and $E_0 = \{e_1, e_2\}$. Without loss of generality, we can assume that $(x, y) \in Q_{n-1}^0$.

Case A. $e_1, e_2 \in Q_{n-1}^0$

Applying Lemma 3, we can construct a Hamiltonian cycle $\langle u, x, y, v \xrightarrow{P(v,u)} u \rangle$ passing through the edges $e_1, e_2, (x, y)$ of Q_3^0 . Applying Lemma 1, we can obtain a Hamiltonian path $P(u^{(1)}, v^{(1)})$ of Q_3^1 . Thus, $\langle u, u^{(1)} \xrightarrow{P(u^{(1)}, v^{(1)})} v^{(1)}, v \xrightarrow{P(v,u)} u \rangle$ forms a Hamiltonian cycle passing e_1, e_2 of $Q_4 - \{x, y\}$.

Case B. $e_1, e_2 \in Q_{n-1}^1$

Applying Lemma 1, we can obtain a Hamiltonian cycle $\langle u, v \xrightarrow{P(v,u)} u \rangle$ of $Q_3^0 - \{x, y\}$ for $(u^{(1)}, v^{(1)}) \notin E_0$. Applying Lemma 3, we can construct a Hamiltonian cycle $\langle u^{(1)} \xrightarrow{P(u^{(1)}, v^{(1)})} v^{(1)}, u^{(1)} \rangle$ passing through the edges $e_1, e_2, (u^{(1)}, v^{(1)})$ of Q_3^1 . Thus, $\langle u, u^{(1)} \xrightarrow{P(u^{(1)}, v^{(1)})} v^{(1)}, v \xrightarrow{P(v,u)} u \rangle$ forms a Hamiltonian cycle passing through e_1, e_2 of $Q_4 - \{x, y\}$.

Case C. e_1 and e_2 are in different subcube

Without loss of generality, we can assume that $e_1 \in Q_3^0$ and $e_2 \in Q_3^1$. Applying Lemma 3, we can construct a Hamiltonian cycle $\langle u, x, y, v \xrightarrow{P(v,u)} u \rangle$ passing through the edges $e_1, (x, y)$ of Q_3^0 . Applying Lemma 4, we can obtain a Hamiltonian path $P(u^{(1)}, v^{(1)})$ passing through the edge e_2 of Q_3^1 . Thus, $\langle u, u^{(1)} \xrightarrow{P(u^{(1)}, v^{(1)})} v^{(1)}, v \xrightarrow{P(v,u)} u \rangle$ forms a Hamiltonian cycle passing through E_0 of Q_4 .

In the follows, we will prove this theorem for $n \geq 5$. Without loss of generality, we can assume that $2|F_a^0| + 2|F_e^0| + |E_0^0| \geq 2|F_a^1| + 2|F_e^1| + |E_0^1|$. We will show this theorem in the following cases.

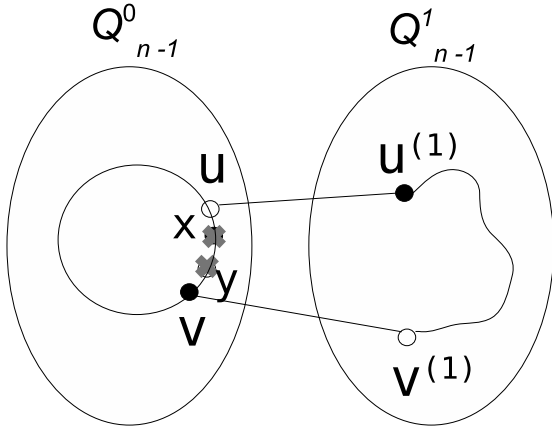


Figure 2: The case1.1 of Theorem 1.

Case1. $2|F_a^0| + 2|F_e^0| + |E_0^0| \geq 2|F_a^1| + 2|F_e^1| + |E_0^1| - 1$

Let (x, y) be a pair of adjacently faulty vertices of F_a for $x \in V_0$ and $y \in V_1$.

Case1.1 $|E_0^0| = |E_0|$

By induction hypothesis, there exists a Hamiltonian cycle C passing through all edges of E_0 of $Q_{n-1}^0 - F_e - (F_a - \{x, y\})$. Suppose that (x, y) is an edge of C . We can denote C as $\langle u, x, y, v \xrightarrow{P(v,u)} u \rangle$. Applying Lemma 1, we can obtain a Hamiltonian path $P(u^{(1)}, v^{(1)})$ of Q_{n-1}^1 . Thus, $\langle u, u^{(1)} \xrightarrow{P(u^{(1)}, v^{(1)})} v^{(1)}, v \xrightarrow{P(v,u)} u \rangle$ forms a Hamiltonian cycle passing through E_0 of $Q_n - F_a - F_e$, as illustrated in Figure 2.

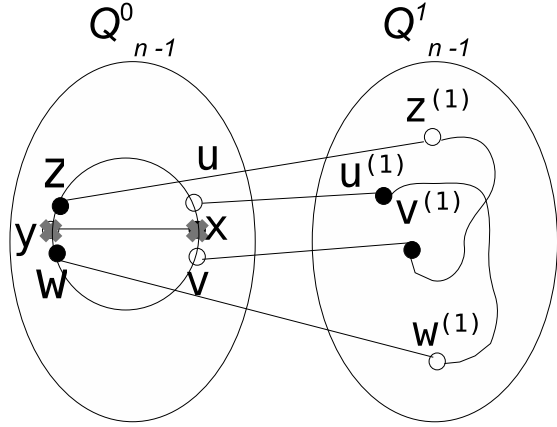


Figure 3: The case1.1 of Theorem 1.

Suppose that (x, y) is not an edge of C . We can denote C as $\langle u, x, v \xrightarrow{P(v,w)} w, y, z \xrightarrow{P(z,u)} u \rangle$. Applying Lemma 2, we can construct two spanning disjoint paths $P(w^{(1)}, u^{(1)})$ and $P(z^{(1)}, v^{(1)})$ of Q_{n-1}^1 . Thus, $\langle v \xrightarrow{P(v,w)} w, w^{(1)} \xrightarrow{P(w^{(1)}, u^{(1)})} u^{(1)}, u \xrightarrow{P(u,z)} z, z^{(1)} \xrightarrow{P(z^{(1)}, v^{(1)})} v^{(1)}, v \rangle$ forms a Hamiltonian cycle passing E_0 of $Q_n - F_a - F_e$. Figure 3 is the illustration for this case.

Case1.2 $|E_0^0| = |E_0| - 1$ and $|E_0^1| = 0$

By induction hypothesis, there exists a Hamiltonian cycle C passing through all edges of $E_0^0 \cup \{(x, y)\}$ of $Q_{n-1}^0 - F_e - (F_a - \{x, y\})$. Let $\{(z, z^{(1)})\} = E_0 - E_0^0$. We can denote C as $\langle u, x, y, v \xrightarrow{P(v,z)} z, w \xrightarrow{P(w,u)} u \rangle$ for $(z, w) \notin E_0$. Applying Lemma 2, we can construct two spanning disjoint paths

$P(u^{(1)}, v^{(1)})$ and $P(z^{(1)}, w^{(1)})$ of Q_{n-1}^1 .
 Thus, $\langle u, u^{(1)} \xrightarrow{P(u^{(1)}, v^{(1)})} v^{(1)}, v \xrightarrow{P(v, z)} z, z^{(1)} \xrightarrow{P(z^{(1)}, w^{(1)})} w^{(1)}, w \xrightarrow{P(w, u)} u \rangle$ forms a Hamiltonian cycle passing through E_0 of $Q_n - F_a - F_e$. Figure 4 is the illustration for this case.

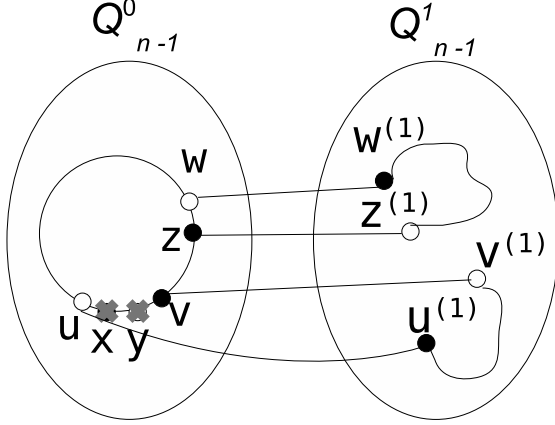


Figure 4: The case1.2 of Theorem 1.

Case1.3 $|E_0^0| = |E_0| - 1$ and $|E_0^1| = 1$

By induction hypothesis, there exists a Hamiltonian cycle $\langle u, x, y, v \xrightarrow{P(v, u)} u \rangle$ passing through all edges of $E_0^0 \cup \{(x, y)\}$ of $Q_{n-1}^0 - F_e - (F_a - \{x, y\})$. Applying Lemma 4, we can construct a Hamiltonian path $P(u^{(1)}, v^{(1)})$ passing through the edge in E_0^1 of Q_{n-1}^1 . Thus, $\langle u, u^{(1)} \xrightarrow{P(u^{(1)}, v^{(1)})} v^{(1)}, v \xrightarrow{P(v, u)} u \rangle$ forms a Hamiltonian cycle passing through E_0 of $Q_n - F_a - F_e$. Figure 5 is the illustration for this case.

Case2 $2|F_a^0| + 2|F_e^0| + |E_0^0| \leq 2|F_a| + 2|F_e| + |E_0| - 2$

Case2.1 $|E_0^1| + |E_0^0| = |E_0| - 1$ and $2|F_a^1| + 2|F_e^1| + |E_0^1| = 1$

Let $\{(z, z^{(1)})\} = E_0 - E_0^0 - E_0^1$. By induction hypothesis, there exists a Hamiltonian cycle $C = \langle z, w \xrightarrow{P(w, z)} z \rangle$ passing through all edges in E_0^0 of $Q_{n-1}^0 - F_e^0 - F_a^0$ for $(z, w), (z^{(1)}, w^{(1)}) \notin E_0$. Suppose that $(z^{(1)}, w^{(1)}) \notin E_0$. Since $2|F_a^0| + 2|F_e^0| + |E_0^0| \geq 2|F_a^1| + 2|F_e^1| + |E_0^1|$ and $n \geq 5$, $2|F_a^1| + 2|F_e^1| + |E_0^1| \leq n - 2 \leq$

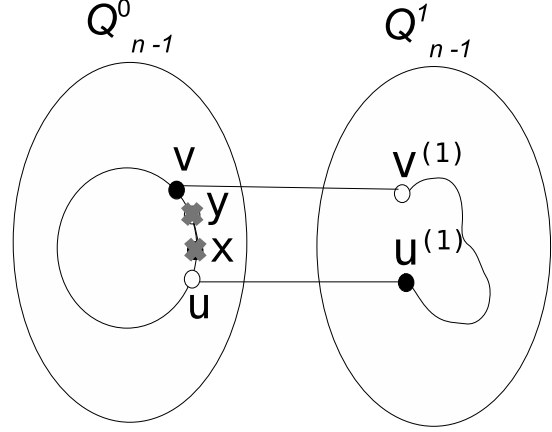


Figure 5: The case1.3 of Theorem 1.

$2n - 7$. Thus, $|E_0^1 \cup \{(z^{(1)}, w^{(1)})\}| \leq 2n - 6 - 2|F_a^1| - 2|F_e^1|$. By induction hypothesis, there exists a Hamiltonian cycle $\langle w^{(1)}, z^{(1)} \xrightarrow{P(z^{(1)}, w^{(1)})} w^{(1)} \rangle$ passing through all edges in $E_0^1 \cup \{(z^{(1)}, w^{(1)})\}$ of $Q_{n-1}^1 - F_e^1 - F_a^1$. Therefore, $\langle z, z^{(1)} \xrightarrow{P(z^{(1)}, w^{(1)})} w^{(1)}, w \xrightarrow{P(w, z)} z \rangle$ is a Hamiltonian cycle passing through E_0 of $Q_n - F_a - F_e$. Suppose that $(z^{(1)}, w^{(1)}) \in E_0$. Let (x, y) be an edge of the Hamiltonian cycle C for $(x, y), (x^{(1)}, y^{(1)}) \notin E_0$. We can denote C as $\langle z, w \xrightarrow{P(w, x)} x, y \xrightarrow{P(y, z)} z \rangle$. Applying Lemma 2, we can construct a Hamiltonian path $P(x^{(1)}, y^{(1)})$ of $Q_{n-1}^1 - \{z^{(1)}, w^{(1)}\}$. Thus, $\langle z, z^{(1)}, w^{(1)}, w \xrightarrow{P(w, x)} x, x^{(1)} \xrightarrow{P(x^{(1)}, y^{(1)})} y^{(1)}, y \xrightarrow{P(y, z)} z \rangle$ is a Hamiltonian cycle passing through E_0 of $Q_n - F_a - F_e$.

Case2.2 $|E_0^1| + |E_0^0| = |E_0| - 1$ and $2|F_a^1| + 2|F_e^1| + |E_0^1| \geq 2$

Let $\{(z, z^{(1)})\} = E_0 - E_0^0 - E_0^1$ for $z \in V_0$. By induction hypothesis, there exists a Hamiltonian cycle $\langle z, w \xrightarrow{P(w, z)} z \rangle$ passing through all edges in E_0^0 of $Q_{n-1}^0 - F_e^0 - F_a^0$ for $(z, w), (z^{(1)}, w^{(1)}) \notin E_0$. Since $2|F_a^0| + 2|F_e^0| + |E_0^0| \geq 2|F_a^1| + 2|F_e^1| + |E_0^1|$ and $n \geq 5$, $2|F_a^1| + 2|F_e^1| + |E_0^1| \leq n - 2 \leq 2n - 7$. Thus, $|E_0^1 \cup \{(z^{(1)}, w^{(1)})\}| \leq 2n - 6 - 2|F_a^1| - 2|F_e^1|$.

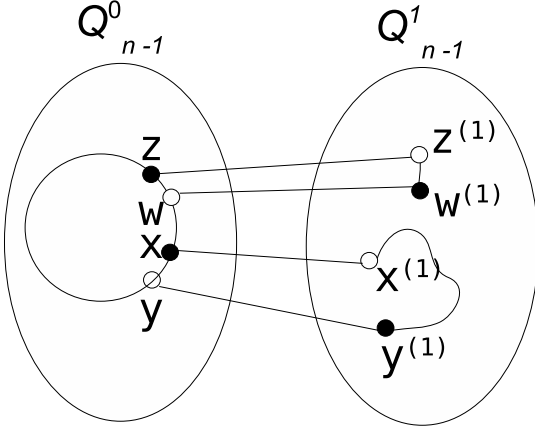


Figure 6: The case 2.1 of Theorem 1.

By induction hypothesis, there exists a Hamiltonian cycle $\langle w^{(1)}, z^{(1)} \xrightarrow{P(z^{(1)}, w^{(1)})} w^{(1)} \rangle$ passing through all edges in $E_0^1 \cup \{(z^{(1)}, w^{(1)})\}$ of $Q_{n-1}^1 - F_e^1 - F_a^1$. Therefore, $\langle z, z^{(1)} \xrightarrow{P(z^{(1)}, w^{(1)})} w^{(1)}, w \xrightarrow{P(w, z)} z \rangle$ is a Hamiltonian cycle passing through E_0 of $Q_n - F_a - F_e$. Figure 7 is the illustration for this case.

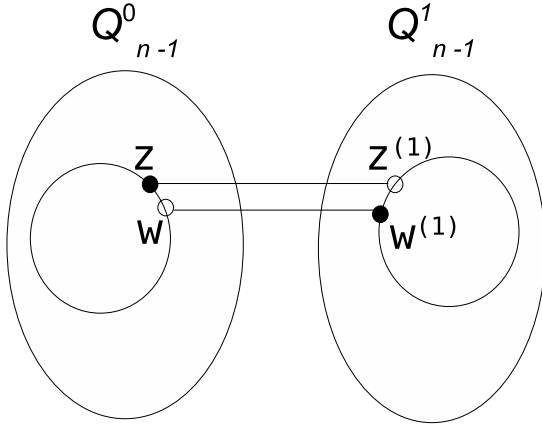


Figure 7: The case 2.2 of Theorem 1.

Case 2.3 $|E_0^1| + |E_0^0| = |E_0|$

By induction hypothesis, there exists a Hamiltonian cycle $\langle u, v \xrightarrow{P(v, u)} u \rangle$ passing through all edges in E_0^0 of $Q_{n-1}^0 -$

$F_e^0 - F_a^0$ for $(u, v), (u^{(1)}, v^{(1)}) \notin E_0$. Since $2|F_a^0| + 2|F_e^0| + |E_0^0| \geq 2|F_a^1| + 2|F_e^1| + |E_0^1|$ and $n \geq 5$, $2|F_a^1| + 2|F_e^1| + |E_0^1| \leq n - 2 \leq 2n - 7$. Thus, $|E_0^1 \cup \{(u^{(1)}, v^{(1)})\}| \leq 2n - 6 - 2|F_a^1| - 2|F_e^1|$. By induction hypothesis, there exists a Hamiltonian cycle $\langle v^{(1)}, u^{(1)} \xrightarrow{P(u^{(1)}, v^{(1)})} v^{(1)} \rangle$ passing through all edges in $E_0^1 \cup \{(v^{(1)}, u^{(1)})\}$ of $Q_{n-1}^1 - F_e^1 - F_a^1$. Therefore, $\langle u, u^{(1)} \xrightarrow{P(u^{(1)}, v^{(1)})} v^{(1)}, v \xrightarrow{P(v, u)} u \rangle$ is a Hamiltonian cycle passing through E_0 of $Q_n - F_a - F_e$. $\square$

4 Conclusion

In this paper, we show that there is a Hamiltonian cycle passing through all edges in E_0 of $Q_n - F_a - F_e$ if $|F_e| + |F_a| \leq n - 2$ and $|E_0| \leq 2n - 4 - 2|F_e| - 2|F_a|$ where the subgraph induced by E_0 is a linear forest. We will further explore the Hamiltonian paths passing through prescribed edges with adjacently faulty vertices.

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